

Summary of “Hidden Markov Neural Networks”

The paper introduces Hidden Markov Neural Networks (HMNNs), a hybrid framework that combines Hidden Markov Models (HMMs) with Bayesian Neural Networks (BNNs). The main motivation is to solve problems in continual learning and time-series forecasting, where data distributions change over time.

Traditional continual learning methods mainly focus on preventing catastrophic forgetting, meaning they try to preserve all previous knowledge. However, in dynamic environments, retaining outdated information can reduce performance. Therefore, the authors argue that intelligent forgetting is equally important as remembering. HMNNs are designed to balance adaptation to new information with controlled forgetting of irrelevant past knowledge.

Main Idea of the Paper

The key contribution of the paper is the idea of modeling neural network weights as hidden states in a Hidden Markov Model. Instead of treating network weights as fixed parameters, HMNNs allow them to evolve over time according to a stochastic transition process.

Core Components

- Neural network weights are treated as latent hidden states.
- Observed data act as emissions in the HMM framework.
- Variational inference is used to approximate posterior distributions.
- Sequential Bayes by Backprop is used for online learning.
- Variational DropConnect is introduced for regularization.

This formulation allows the model to continuously update itself as new data arrive while maintaining uncertainty estimates over the parameters.

Hidden Markov Neural Network Structure

In HMNNs:

- The hidden process represents the evolution of neural network weights over time.
- The observed process corresponds to the training data available at each time step.
- A transition kernel determines how weights evolve from one step to another.
- The emission distribution models the likelihood of the observed data given the current weights.

The authors specifically use a Factorial Hidden Markov Model (FHMM), where weights evolve independently. This factorization reduces computational complexity and makes inference more scalable.

Variational Inference

Exact Bayesian inference is intractable for neural networks, especially when weights evolve over time. To solve this, the paper applies variational inference.

The filtering distribution over the weights is approximated sequentially using a variational distribution. At each time step:

1. Previous posterior estimates are propagated through the transition kernel.
2. New observations update the posterior.
3. Optimization minimizes the KL-divergence between the approximation and the true posterior.

The method effectively becomes a sequential extension of Bayes by Backprop.

Sequential Reparameterization Trick

The authors reformulate the reparameterization trick for sequential learning. Monte Carlo sampling is used to estimate gradients, enabling efficient optimization through gradient descent.

Variational DropConnect

One of the major contributions of the paper is variational DropConnect.

Purpose

Variational DropConnect acts as a regularization method similar to DropOut and DropConnect but within a Bayesian framework.

Key Idea

The variational posterior is modeled as a mixture of Gaussian distributions:

- Some weights are encouraged toward zero.
- Other weights remain active with learned means and variances.

This improves robustness and prevents overfitting while maintaining uncertainty estimation.

The experiments show that variational DropConnect performs better than standard Bayes by Backprop on MNIST classification tasks.

Experiments and Results

The paper evaluates HMNNs on multiple tasks to test adaptation, uncertainty estimation, and predictive performance.

Experiments Conducted

- MNIST classification
- Dynamic “two moons” dataset
- Concept drift logistic regression
- Evolving MNIST classifier
- Video next-frame prediction (waving flag dataset)

Baselines Compared Against

- Bayes by Backprop (BBP)
- Elastic Weight Consolidation (EWC)
- Variational Continual Learning (VCL)
- LSTM networks
- Bayesian adaptation methods

Main Findings

- HMNNs achieved competitive or superior performance in dynamic environments.
- Variational DropConnect improved classification accuracy on MNIST.
- HMNNs successfully handled concept drift by adapting to changing data distributions.
- The framework provided strong uncertainty quantification through posterior sampling.
- In video prediction tasks, HMNNs produced more reliable predictions than LSTMs and sequential DropConnect.

